REZA SHAMJI

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EDUCATION

Harvard University: A.B. in Computer Science (Secondary in Economics). GPA 3.71 *May 2025*
Relevant Coursework: Algorithms & Limitations; Distributed Systems & Machine Organization; Semantics of PL; Engineering Usable Interactive Systems; Probability; Linear Algebra; Discrete Math; Multivariable Calculus

PREPRINTS & MANUSCRIPTS

"Democratizing AI Scientists using ToolUniverse." *arXiv, 2025*. Gao, S., Zhu, R., SuiC, P., Kong, Z., Aldogom, S., Huang, Y., Noori, A., Shamji, R., et al. (*Nature Biotechnology submission under review).* Co-author — conducted during collaboration with Zitnik Lab to develop an open ecosystem (aiscientist.tools) enabling agent access to 600+ scientific tools; highlighted in *Nature* ("How AI Agents Will Change Research") and *Decoding Bio's BioByte* ("ToolUniverse Democratizes AI Scientists").

"Understanding the Design Space and Cross-Modality Transfer for Vision-Language Models." (*ICLR 2026 submission under review)*. Co-author — conducted during Kempner Institute internship under Yilun Du and Sham Kakade on multimodal VLM architectures (manuscript available upon request)

RESEARCH EXPERIENCE

Research Engineer *Sept 2025-Present*
In collaboration with the Zitnik Lab, Harvard Medical School
- Hybrid keyword + embedding datastore (SQLite FTS5, FAISS): Built pipeline (ingest, deduplicate, embed, index) with hybrid search via a simple CLI and agent API; added Hugging Face sync and local caching. Enables both non-technical labs and autonomous agents to turn raw documents into shared knowledge bases that any team's agents can query immediately through ToolUniverse.
- EUHealth (public-health datasets): Developed tools that transform FAIR DCAT-AP metadata from the EU Health Information Portal into searchable, weekly-refreshed index with topic/country/language filters and a link classifier that flags pages with downloadable data. This enables automated discovery of population-level datasets by agents for reproducible epidemiology and policy analyses.
- AlphaFold (protein structures): Integrated DeepMind/EMBL-EBI AlphaFold via tools with a UniProt-based flow so agents can retrieve predicted 3D structures, pLDDT, and PDB/mmCIF files. Adds structural biology evidence to agentic reasoning pipelines.
- ODPHP (preventive care): Integrated U.S. HHS MyHealthFinder API to structure demographic-specific (age/sex/pregnancy) preventative-care recommendations, enabling agents to generate step-by-step healthcare guidance grounded in official U.S. policy.

Machine Learning Research Engineer Intern (Multimodal AI Team) *Jun-Sept 2025*
Kempner Institute for Natural & Artificial Intelligence, Harvard University
- Built end-to-end VQA benchmarking (ChartQA, RealWorldQA, MMT-Bench, MathVista, DocVQA, TextVQA): dataset factories, OCR/table serialization, collate functions, and benchmark-specific scorers (numeric relative-error, ANLS, MCQ), enabling rigorous multimodal evaluation. (Repository private; code excerpts available upon request)
- Engineered transformer infrastructure for multimodal VLMs: implemented configurable query-key normalization (LayerNorm, RMSNorm, custom) to stabilize OCR/vision token processing, integrated Queen-38B into cross-attention fusion with selective freezing, and developed a YAML-driven learning rate scheduler registry (cosine warmup, custom schedulers).
- Ran multi-node training/eval (FSDP/DDP on H100s) using Slurm with W&B monitoring

Projects: **ChainEnv RL Benchmarks (JAX)** — laptop-friendly 1-D chain to study exploration; compares PPO/PQN/DDPG/SAC. Key finding: performance becomes exploration-limited as difficulty rises; Q(λ) accelerates credit after first success. Github Repo: /jax-chainenv-benchmarks
Technical Skills: Python (PyTorch; HF Transformers/Tokenizers; TorchVision), Distributed (DDP/FSDP/DTensor), Slurm/NCCL, WebDataset, Pandas/Numpy, W&B, YAML; C++, Java, OCaml; Docker/Conda; SQL • Hardware: multi-node H100/A100 (e.g. 4×4 GPUs, FSDP)

INDUSTRY EXPERIENCE

Summer Business Analyst *Jun-Aug 2024*
McKinsey & Company
- Led firm-wide GenAI enablement plan (capability map, LLM use-case playbooks) approved by the CEO.
- Owned a supply-chain workstream for a power manufacturer (supplier RFP analysis, negotiation support, ops dashboards), contributing to $15m savings in a $200M program.

Quantitative ESG Intern, Research Analyst *Jun-Aug 2023*
Los Angeles Capital Management
- Built Python/Spark pipelines joining Bloomberg, MSCI, and Goldman Sachs datasets for large-scale ESG signal generation and conducted implied temperature rise analysis for climate-risk profiling and portfolio reporting.
- Developed and back-tested a core–satellite ESG factor model integrated into the stock-selection system

LEADERSHIP & SERVICE

Harvard Class Program Marshal '25; South Asian Association Social Chair; Harvard Club Soccer & Golf; Freshman Outdoor Program Leader.
- Aga Khan Foundation (2015–present): Designed curriculum and led a week-long youth program (130 participants) on mental health and identity; hired and trained 60 staff; managed $50k budget.
- AAPI Outreach Intern of Biden Coalition, TX (2020–2021): Coordinated bilingual outreach strategy; contributed to a 260% increase in AAPI turnout and 9 of 12 congressional victories.

LANGUAGES: English (native); Mandarin (fluent)